

Jackson Bopp

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Computer Engineering student seeking an entry-level engineering role spanning embedded systems, hardware, and software, graduating Fall 2026.

Skills

Languages: C, C++, Python, TypeScript, Verilog/VHDL, RISC-V Assembly, Bash, MATLAB

Hardware & Protocols: MSP430FR2355, ESP32, STM32, RP2040, FPGA (Xilinx), Raspberry Pi, Arduino, UART, I2C, SPI, PWM, MQTT, GPIO, Bluetooth LE

Tools & Infrastructure: Git, Docker, GitHub Actions, Gerrit code review, Linux, GDB, Xilinx ISE, PostgreSQL

Certifications: Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

Education

University of South Florida

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.52/4.00

Tampa, FL

Graduating Fall 2026

Projects

FPGA Audio Message Recorder | Xilinx Spartan-6, Verilog, PicoBlaze, DDR2 SDRAM, I2C, UART

Spring 2026

- Collaborated with a 3-member team to design, synthesize, and deploy a full audio recording and playback system on a Xilinx Spartan-6 FPGA.
- Wrote the top-level Verilog FSM managing record, play, pause, rewind, and a 4-message library with independent delete, backed by a DDR2 SDRAM interface for sample storage and an I2C master driving the SSM2603 audio codec.
- Programmed a PicoBlaze embedded soft-core in assembly for the UART menu and debugged real hardware issues (audio distortion, a rewind pointer bug) across several controller revisions to reach a stable final build.

Testbench | Python, MQTT, Docker, GitHub Actions, MSP430FR2355, ESP32, STM32F103C8, RP2040

July 2026

- Designed a hardware-agnostic test automation platform streaming live telemetry from four different microcontrollers through a Raspberry Pi bridge over MQTT to a web dashboard.
- Implemented reference firmware for the MSP430FR2355, ESP32, STM32F103C8, and RP2040 emitting a shared JSON-over-UART frame format, plus the Python MQTT bridge and a hardware simulator for running demos and regression tests without physical hardware.
- Set up a GitHub Actions workflow that spins up Postgres and Mosquitto containers and runs an end-to-end simulated-hardware regression test on every push.

Radiology Second-Opinion Agent | FastAPI, Docker Compose, MLflow, Evidently, pytest

July 2026

- Data and MLOps engineer on a 4-person hackathon team; owned the model serving layer and its test coverage, including unit tests, mock-LLM tests, and endpoint validation, plus Evidently-based data drift monitoring in production.

Open Source

Open Source Contributions | C++, Go, Python, C, D-Bus, Linux, Gerrit and GitHub code review

August 2026

- OpenBMC phosphor-user-manager:** diagnosed and fixed two bugs in a BMC user management daemon, an unhandled exception that could abort account setup for every user over one bad record, and a C++ construction-order bug that let unit tests silently run against the real system instead of their mocks; verified with full before-and-after test counts.
- Zephyr RTOS:** found and fixed a devicetree macro bug in the MIPI DBI display driver that produced an incorrect chip select GPIO check for out-of-range devices; submitted with a compile-time regression test.
- TinyGo:** identified an unsynchronized write racing against every other lock-protected access to the same runtime data, built the compiler from source to verify, and submitted the fix.
- PlatformIO:** root-caused and fixed a crash in the serial device monitor triggered by non-interactive input; merged upstream with an added regression test.

Experience

City of Winter Haven Technology Services

Smart City Student Volunteer

April 2026 - Present

Winter Haven, FL

- Installed server hardware and performed rack maintenance for city technology infrastructure.
- Provisioned and configured YubiKey security keys for staff authentication.
- Resolved IT support tickets and optimized PyTorch/TensorFlow environments for the city's early-stage AI hosting infrastructure.

Subs N' Such

Manager, Shift Leader, and Front-End Web Support

April 2023 - March 2026

Lutz, FL

- Developed the front-end of the restaurant's Square-based ordering site, working within the platform's UI constraints to deliver a functional ordering experience.
- Managed shift operations and enforced food safety procedures.

Involvement

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE